MAXIMAL SLICING OF A COHEN-MACAULAY GROUPOID SCHEME AT A FINITE TYPE POINT, KEEPING H, S' AND T' COHEN-MACAULAY AND LOCALLY OF FINITE PRESENTATION

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Allowed standard prerequisites (editorial).

- Definition of a finite type point of a scheme (morphisms chapter, Definition morphisms-definition-finite-type-point, canonical tag 02J1). cited at statement lines 2251-2252. Fixes the class of points u of U at which the slicing problem is posed.
- The slicing situation: notation F__u, G__u, G the stabilizer group scheme, and the objects F'__u, G'__u, R', s', t', h of a restriction (more-groupoids chapter, Situation situation-slice, canonical tag 04MY; kind reference). cited at statement lines 2253-2254. Supplies the notation in which the conclusion is phrased: the fibres and their dimensions, the stabilizer dimension dim(G'__u), and the morphisms s', t' and h of the restricted groupoid.
- The slice lemma: one slicing step at u giving an immersion g with properties (1), (2), (3), a strict orbit dimension drop, and h Cohen-Macaulay at (u, e(u)) (more-groupoids chapter, Lemma lemma-slice, canonical tag 0461). cited at proof lines 2277-2285. The workhorse of the induction step: produces the single slice whose h is Cohen-Macaulay at (u, e(u)) and whose fibre dimension strictly drops.
- Remark producing the largest open on which a morphism locally of finite presentation is Cohen-Macaulay (more-groupoids chapter, Remark remark-local-source-apply, canonical tag 047W). cited at proof lines 2285-2290. Turns Cohen-Macaulayness of h at the single point (u, e(u)) into Cohen-Macaulayness of h on all of U", by shrinking to the largest such open, which contains u.
- Restriction lemma transferring properties of h to s' and t' of the restricted groupoid (more-groupoids chapter, Lemma lemma-restrict-property, canonical tag 04MV). cited at proof lines 2293-2295. Concludes that s' and t' are locally of finite presentation and Cohen-Macaulay once h is, via the two base change lemmas.

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- Base change of morphisms locally of finite presentation (morphisms chapter, Lemma morphisms-lemma-base-change-finite-presentation, canonical tag 01TS). cited at proof line 2296. Input to the restrict-property application: local finite presentation of s and t passes to the fibre products over which s' and t' factor.
- Base change of Cohen-Macaulay morphisms (more-morphisms chapter, Lemma more-morphisms-lemma-base-change-CM, canonical tag 045T). cited at proof lines 2297-2298. Input to the restrict-property application: Cohen-Macaulayness of h passes to the base changes over which s' and t' factor.
- Identification of finite type points under a locally finite presentation morphism (morphisms chapter, Lemma morphisms-lemma-identify-finite-type-points, canonical tag 02J2). cited at proof lines 2303-2306. Shows u remains a finite type point of U' after the immersion g, so the induction hypothesis applies to the restricted groupoid at u.
- Double restriction lemma: the composition h" for a composed slicing stays locally of finite presentation and Cohen-Macaulay (more-groupoids chapter, Lemma lemma-double-restrict-property, canonical tag 04MX). cited at proof lines 2318-2319. Verifies property (4) for the composed solution g" = g composed with g'.
- Cohen-Macaulay morphisms are fppf local on the target (more-morphisms chapter, Lemma more-morphisms-lemma-CM-local-source-and-target, canonical tag 045V). cited at proof lines 2321-2322. Used inside the double restrict-property application to descend Cohen-Macaulayness along the fppf covering of the target.

Source reference. Original definition 02J1: sources/stacks-chow/tags.txt, source lines 3320–3320. Cited at statement lines 2251-2252: u is taken to be a finite type point of U, the class of points at which the slicing problem is posed.

Source reference. Original reference 04MY: sources/stacks-chow/tags.txt, source lines 6059–6059. Cited at statement lines 2253-2254: notation as in Situation slice supplies F_u, G_u, the stabilizer group scheme G, and the restriction objects F'_u, G'_u, R', s', t' in which the six listed properties are phrased.

Source reference. Original lemma 0461: sources/stacks-chow/tags.txt, source lines 5456–5456. Cited at proof lines 2277-2285: the induction step slices once at u, getting the immersion g with properties (1), (2), (3), the strict orbit dimension drop, and h Cohen-Macaulay at (u, e(u)).

Source reference. Original remark 047W: sources/stacks-chow/tags.txt, source lines 5523–5523. Cited at proof lines 2285-2290: produces the largest open U" of U' on which h is Cohen-Macaulay, containing (u, e(u)) and hence u, after which h is Cohen-Macaulay everywhere.

Source reference. Original lemma 04MV: sources/stacks-chow/tags.txt, source lines 6056–6056. Cited at proof lines 2293-2295: transfers local finite presentation and Cohen-Macaulayness from h to s' and t' of the restriction.

Source reference. Original lemma 01TS: sources/stacks-chow/tags.txt, source lines 2385–2385. Cited at proof line 2296 inside the restrict-property application: local finite presentation passes along base change.

Source reference. Original lemma 045T: sources/stacks-chow/tags.txt, source lines 5448–5448. Cited at proof lines 2297-2298 inside the restrict-property application: Cohen-Macaulayness passes along base change.

Source reference. Original lemma 02J2: sources/stacks-chow/tags.txt, source lines 3321–3321. Cited at proof lines 2303-2306: u is still a finite type point of U' after the immersion g, so the induction hypothesis applies at u.

Source reference. Original lemma 04MX: sources/stacks-chow/tags.txt, source lines 6058–6058. Cited at proof lines 2318-2319: h" for the composed slicing is locally of finite presentation and Cohen-Macaulay, giving property (4) of the composed solution.

Source reference. Original lemma 045V: sources/stacks-chow/tags.txt, source lines 5450–5450. Cited at proof lines 2321-2322 inside the double restrict-property application: Cohen-Macaulay morphisms are fppf local on the target.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c, e, i) be a groupoid scheme over S. Let $G \to U$ be the stabilizer group scheme. Assume s and t are Cohen-Macaulay and locally of finite presentation. Let $u \in U$ be a finite type point of the scheme U, see Morphisms, Definition 02J1. With notation as in Situation 04MY there exist an affine scheme U' and a morphism $g : U' \to U$ such that*

(1) *g is an immersion,*
(2) *$u \in U'$,*
(3) *g is locally of finite presentation,*
(4) *the morphism $h : U' \times_{g,U,t} R \longrightarrow U$ is Cohen-Macaulay and locally of finite presentation,*
(5) *the morphisms $s', t' : R' \to U'$ are Cohen-Macaulay and locally of finite presentation, and*
(6) $\dim_{e(u)}(F'_u) = \dim(G'_u)$.

Proof. As s is locally of finite presentation the scheme F_u is locally of finite type over $\kappa(u)$. Hence $\dim_{e(u)}(F_u) < \infty$ and we may argue by induction on $\dim_{e(u)}(F_u)$.

If $\dim_{e(u)}(F_u) = \dim(G_u)$ there is nothing to prove. Assume $\dim_{e(u)}(F_u) > \dim(G_u)$. This means that Lemma 0461 applies and we find a morphism $g : U' \to U$ which has properties (1), (2), (3), instead of (6) we have $\dim_{e(u)}(F'_u) < \dim_{e(u)}(F_u)$, and instead of (4) and (5) we have that the composition

$$h = s \circ \mathrm{pr}_1 : U' \times_{g,U,t} R \longrightarrow U$$

is Cohen-Macaulay at the point $(u, e(u))$. We apply Remark 047W and we obtain an open subscheme $U'' \subset U'$ such that $U'' \times_{g,U,t} R \subset U' \times_{g,U,t} R$ is the largest open subscheme on which h is Cohen-Macaulay. Since $(u, e(u)) \in U'' \times_{g,U,t} R$ we see that $u \in U''$. Hence we may replace U' by U'' and assume that in fact h is Cohen-Macaulay everywhere! By Lemma 04MV we conclude that s', t' are locally of finite presentation and Cohen-Macaulay (use Morphisms, Lemma 01TS and More on Morphisms, Lemma 045T).

By construction $\dim_{e'(u)}(F'_u) < \dim_{e(u)}(F_u)$, so we may apply the induction hypothesis to (U', R', s', t', c') and the point $u \in U'$. Note that u is also a finite type point of U' (for example you can see this using the characterization of finite type points from Morphisms, Lemma 02J2). Let $g' : U'' \to U'$ and $(U'', R'', s'', t'', c'')$

be the solution of the corresponding problem starting with (U', R', s', t', c') and the point $u \in U'$. We claim that the composition

$$g'' = g \circ g' : U'' \longrightarrow U$$

is a solution for the original problem. Properties (1), (2), (3), (5), and (6) are immediate. To see (4) note that the morphism

$$h'' = s \circ \mathrm{pr}_1 : U'' \times_{g'',U,t} R \longrightarrow U$$

is locally of finite presentation and Cohen-Macaulay by an application of Lemma 04MX (use More on Morphisms, Lemma 045V to see that Cohen-Macaulay morphisms are fppf local on the target). □